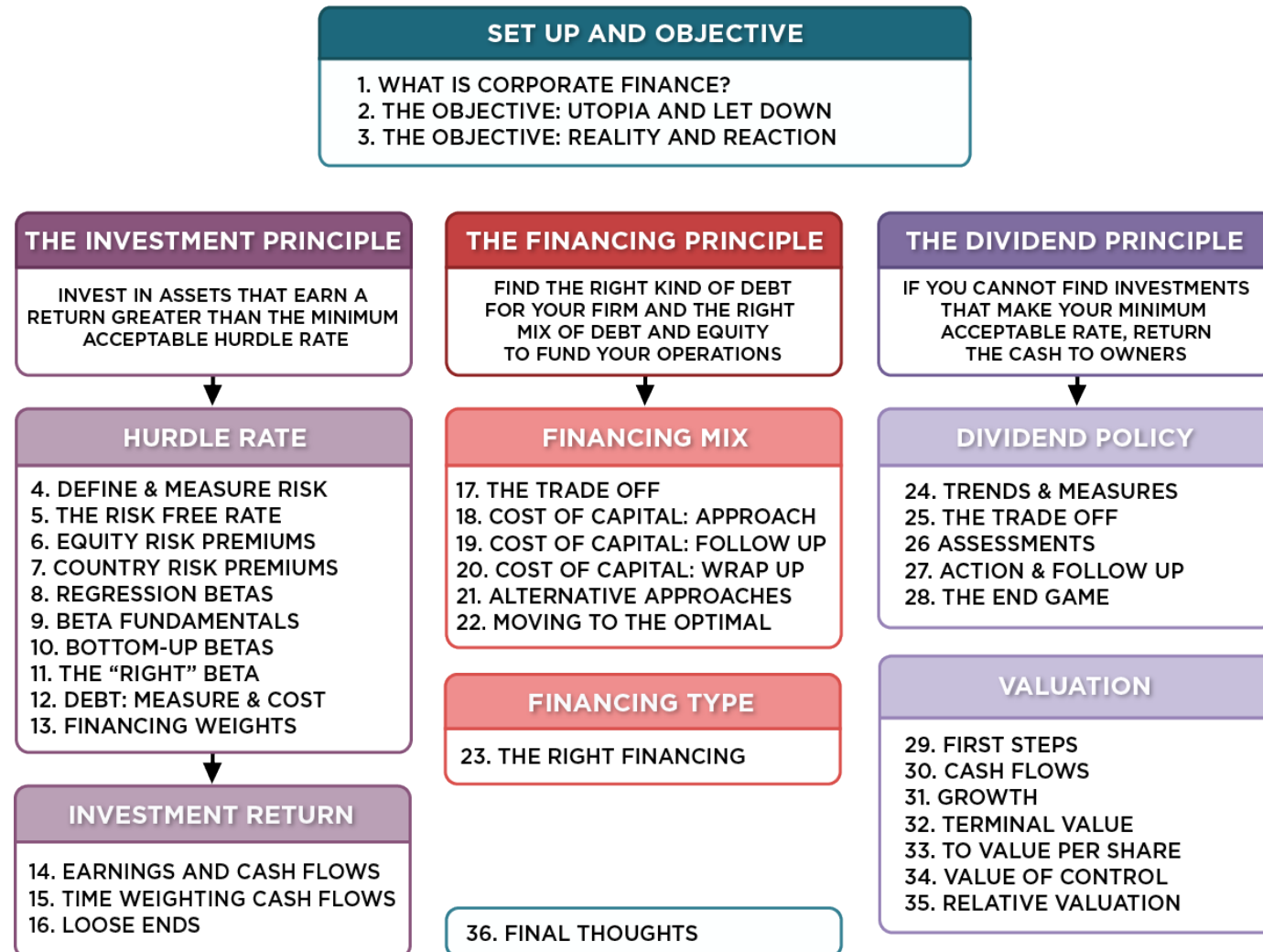


SESSION 11: THE “RIGHT” BETA

The law of large numbers is your best friend.

SESSION 11: THE “RIGHT” BETA



ESTIMATING BOTTOM UP BETAS & COSTS OF EQUITY: VALE

ESTIMATING BOTTOM UP BETAS & COSTS OF EQUITY VALE

	BUSINESS	SAMPLE	SAMPLE SIZE	UNLEVERED BETA OF BUSINESS	REVENUES	PEER GROUP EV/SALES	VALUE OF BUSINESS	PROPORTION OF VALE
	METALS & MINING	GLOBAL FIRMS & MINING MARKET CAP > \$1 BILLION	48	0.86	\$9,013	1.97	\$17,739	16.65%
	IRON ORE	GLOBAL FIRMS IN IRON ORE	78	0.83	\$32,717	2.48	\$81,188	76.20%
	FERTILIZERS	GLOBAL SPECIALITY CHEMICAL FIRMS	693	0.99	\$3,777	1.52	\$5,741	5.39%
	LOGISTICS	GLOBAL TRANSPORTATION FIRMS	223	0.75	\$1,644	1.14	\$1,874	1.76%
	VALE OPERATIONS			0.8440	\$47,151		\$106,543	100%

	BUSINESS	UNLEVERED BETA	D/E RATIO	LEVERED BETA	RISK FREE RATE	ERP	COST OF EQUITY
	METALS & MINING	0.86	54.99%	1.1657	2.75%	7.38%	11.35%
	IRON ORE	0.83	54.99%	1.1358	2.75%	7.38%	11.13%
	FERTILIZERS	0.99	54.99%	1.3493	2.75%	7.38%	12.70%
	LOGISTICS	0.75	54.99%	1.0222	2.75%	7.38%	10.29%
	VALE OPERATIONS	0.84	54.99%	1.1503	2.75%	7.38%	11.23%



VALE: COST OF EQUITY CALCULATION – IN NOMINAL \$R

- To convert a discount rate in one currency to another, all you need are expected inflation rates in the two currencies.

$$(1 + \$ \text{ COST OF EQUITY}) \frac{(1 + \text{INFLATION RATE}_{\text{BRAZIL}})}{(1 + \text{INFLATION RATE}_{\text{US}})} - 1$$

- From US \$ to R\$: If we use 2% as the inflation rate in US dollars and 9% as the inflation ratio in Brazil, we can convert Vale's US dollar cost of equity of 11.23% to a \$R cost of equity:

$$\text{COST OF EQUITY}_{\text{NOMINAL R\$}} = (1 + \$ \text{ COST OF EQUITY}) \frac{(1 + \text{INFLATION RATE}_{\text{BRAZIL}})}{(1 + \text{INFLATION RATE}_{\text{US}})} - 1$$

$$= (1.123) \frac{(1.09)}{(1.02)} - 1 = 18.87\%$$

- Alternatively, you can compute a cost of equity, starting with the \$R riskfree rate of 10.18%.

$$\text{Cost of Equity in \$R} = 10.18\% + 1.15 (7.38\%) = 18.67\%$$

BOTTOM UP BETAS & COSTS OF EQUITY: TATA MOTORS & BAIDU

- Tata Motors: We estimated an unlevered beta of 0.8601 across 76 publicly traded automotive companies (globally) and estimated a levered beta based on Tata Motor's D/E ratio of 41.41% and a marginal tax rate of 32.45% for India:

Levered Beta for Tata Motors = $0.8601 (1 + (1 - .3245) (.4141)) = 1.1007$

Cost of equity (Rs) = $6.57\% + 1.1007 (7.19\%) = 14.49\%$

- Baidu: To estimate its beta, we looked at 42 global companies that derive all or most of their revenues from online advertising and estimated an unlevered beta of 1.30 for the business. Incorporating Baidu's current market debt to equity ratio of 5.23% and the marginal tax rate for China of 25%, we estimate Baidu's current levered beta to be 1.3560.

Levered Beta for Baidu = $1.30 (1 + (1 - .25) (.0523)) = 1.356$

Cost of Equity for Baidu (Renmimbi) = $3.50\% + 1.356 (6.94\%) = 12.91\%$

BOTTOM UP BETAS AND COSTS OF EQUITY: DEUTSCHE BANK

- We break Deutsche Bank down into two businesses – commercial and investment banking.
- We do not unlever or relever betas, because estimating debt and equity for banks is an exercise in futility.

ESTIMATING BOTTOM UP BETAS & COSTS OF EQUITY
Deutsche Bank

BUSINESS	SAMPLE	SAMPLE SIZE	MEDIAN LEVERED BETA	DUETSCHNE NET REVENUES 2012	PROPORTION
BANKING	EUROPEAN DIVERSIFIED BANKS	84	1.0665	19,019 MIL €	54.86%
INVESTMENT BANKING	GLOBAL INVESTMENT BANKS	58	1.2550	15,648 MIL €	45.14%
DEUTSCHE BANK			1.1516	34,667 MIL €	

BUSINESS	BETA	COST OF EQUITY
BANKING	1.0665	$1.75\% + 1.0665 (6.12) = 8.28\%$
INVESTMENT BANKING	1.2550	$1.75\% + 1.2550 (6.12) = 9.44\%$
DEUTSCHE BANK	1.1516	$1.75\% + 1.1516 (6.12\%) = 8.80\%$

ESTIMATING BETAS FOR NON-TRADED ASSETS

- The conventional approaches of estimating betas from regressions do not work for assets that are not traded. There are no stock prices or historical returns that can be used to compute regression betas.
- There are two ways in which betas can be estimated for non-traded assets
 - Using comparable firms
 - Using accounting earnings

USING COMPARABLE FIRMS TO ESTIMATE BETA FOR BOOKSCAPE

<i>Company Name</i>	<i>Industry</i>	<i>Market Capitalization</i>	<i>Levered Beta</i>	<i>Marginal tax rate</i>	<i>Gross D/E ratio</i>	<i>Cash/Firm Value</i>	<i>R²</i>
Red Giant Entertainment	Publishing	\$2.13	0.69	40.00%	0.00%	0.05%	0.1300
CTM Media Holdings	Publishing	\$25.20	1.04	40.00%	17.83%	33.68%	0.1800
Books-A-Million	Book Stores	\$38.60	1.42	40.00%	556.55%	4.14%	0.1900
Dex Media	Publishing	\$90.50	4.92	40.00%	3190.39%	7.86%	0.2200
Martha Stewart Living	Publishing	\$187.70	1.11	40.00%	19.89%	15.86%	0.3500
Barnes & Noble	Book Stores	\$939.30	0.11	40.00%	164.54%	3.22%	0.2600
Scholastic Corporation	Publishing	\$953.80	1.08	40.00%	21.41%	1.36%	0.2750
John Wiley	Publishing	\$2,931.40	0.81	40.00%	29.58%	5.00%	0.3150
Washington Post	Publishing	\$4,833.20	0.68	40.00%	21.04%	16.04%	0.2680
News Corporation	Publishing	\$10,280.40	0.49	40.00%	8.73%	24.05%	0.2300
Thomson Reuters	Publishing	\$31,653.80	0.62	40.00%	26.38%	1.68%	0.2680
Average			1.1796	40.00%	368.76%	10.27%	0.2442
Median			0.8130	40.00%	21.41%	5.00%	0.2600

Unlevered beta for book company = $0.8130 / (1 + (1 - .4) (.2141)) = 0.7205$

Unlevered beta for book business = $0.7205 / (1 - .05) = 0.7584$

ESTIMATING BOOKSCAPE LEVERED BETA AND COST OF EQUITY

- Because the debt/equity ratios used in computing levered betas are market debt equity ratios, and the only debt equity ratio we can compute for Bookscape is a book value debt equity ratio, we have assumed that Bookscape is close to the book industry median market debt to equity ratio of 21.41 percent.
- Using a marginal tax rate of 40 percent for Bookscape, we get a levered beta of 0.8558.

Levered beta for Bookscape = $0.7584[1 + (1 - 0.40)(0.2141)] = 0.8558$

- Using a riskfree rate of 2.75% (US treasury bond rate) and an equity risk premium of 5.5%:

Cost of Equity = $2.75\% + 0.8558(5.5\%) = 7.46\%$

IS BETA AN ADEQUATE MEASURE OF RISK FOR A PRIVATE FIRM?

- Beta measures the risk added on to a diversified portfolio. The owners of most private firms are not diversified. Therefore, using beta to arrive at a cost of equity for a private firm will
 - a. Under estimate the cost of equity for the private firm
 - b. Over estimate the cost of equity for the private firm
 - c. Could under or over estimate the cost of equity for the private firm

TOTAL RISK VERSUS MARKET RISK

- Adjust the beta to reflect total risk rather than market risk. This adjustment is a relatively simple one, since the R squared of the regression measures the proportion of the risk that is market risk.
 - Total Beta = Market Beta / Correlation of the sector with the market
- In the Bookscape example, where the market beta is 0.8558 and the average R-squared of the comparable publicly traded firms is 26.00%; the correlation with the market is 50.99%.

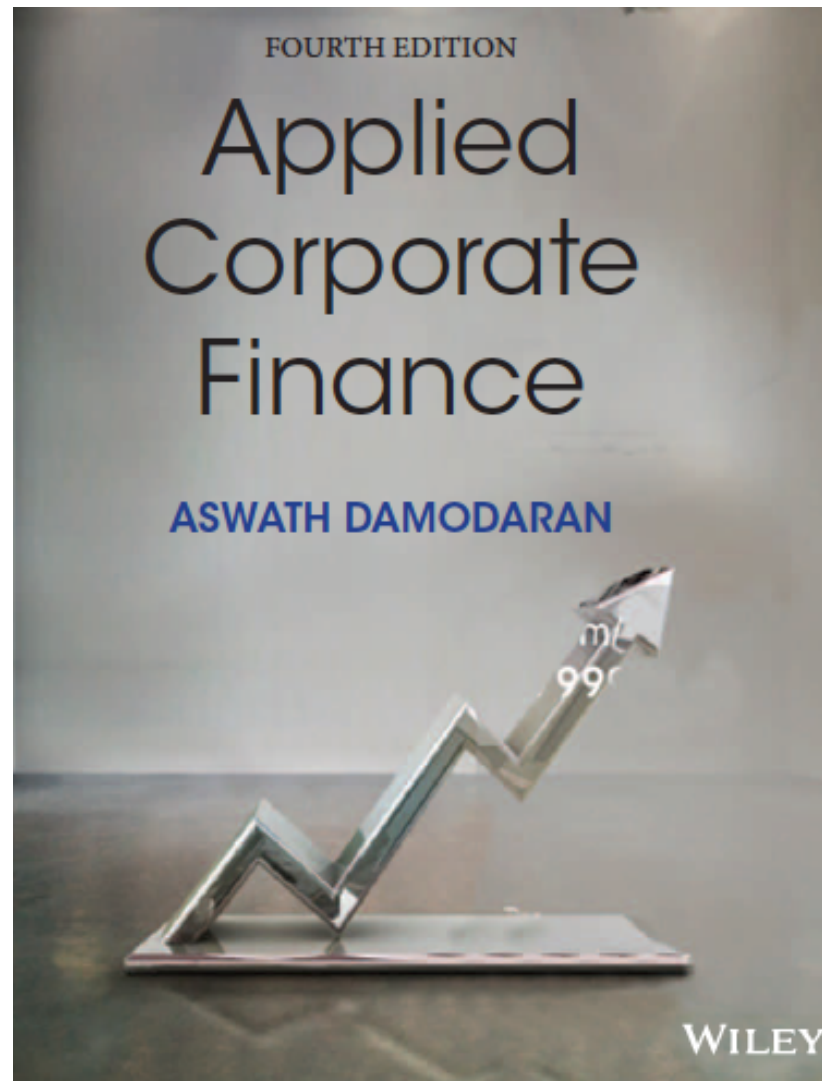
$$\frac{\text{MARKET BETA}}{\sqrt{\text{R SQUARED}}} = \frac{0.8558}{.5099} = 1.6783\%$$

- Total Cost of Equity = 2.75 + 1.6783 (5.5%) = 11.98%

APPLICATION TEST: ESTIMATING A BOTTOM-UP BETA

- Based upon the business or businesses that your firm is in right now, and its current financial leverage, estimate the bottom-up unlevered beta for your firm.
- Data Source: You can get a listing of unlevered betas by industry on my web site by going to updated data.

Task
Estimate the beta
your company
would have, if it
were a private
business.



**Optional:
Read
Chapter 4**